

Final Project Report: Visual-Inertial Navigation using Error-State Extended Kalman Filter (ES-EKF)

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Abstract—In this study we present a Visual Inertial Navigation framework for a planar robot using an Error-State Extended Kalman Filter (ES-EKF). The prediction is propagated with the high rate IMU measurements while the correction updates are incorporated with low-rate visual observations of known landmarks. The visual observations are obtained from a monocular camera and they are processed with color-based segmentation of landmarks with classical image processing techniques. Zero Velocity Update (ZUPT) is also used in this study to reduce the drift of the IMU. The framework is implemented with ROS 2 Humble and Gazebo Fortress. Detailed analyses of the framework is presented in various scenarios, showing the visual measurements’ performance boost compared to dead reckoning.

I. INTRODUCTION

A. Visual Measurement

Accurate state estimation which includes robot pose and orientation is a fundamental requirement for robotics. For visual algorithms such as ORB-SLAM [6], it is possible to provide accurate relative motion estimation using camera’s trajectory and the 3D map structure. This is core mechanism of “Feature Based” SLAM which utilizes The Observation of specific features as in pixel coordinates and The Projection where the predicted pixel coordinates obtained by projecting the corresponding 3D map points onto the camera’s image plane considering current estimate pose of camera. ORB works by minimizing the reprojection errors thus adjusting both camera poses and the 3D point locations to find most consistent solution. For monocular setup with single pinhole camera model, idea suffers from scale ambiguity performing minimization on latent space projection. Robust cost functions including from standard least-squares minimization to Huber Kernel (ρ_{Hub}) optimization of solving $\sum \rho_{Hub}(\|r_{ij}\|_{\sum_{ij}}^2)$ introduces computational overhead. Also robot environments not always carry semantic textures to work on. In a recent study Kannapiran et. al 2021 [7] proposes utilization of RGB-D camera especially for feature poor environments. They extracted 3D vectors and Planes from point cloud and combined data with RGB feature extractors. For feature extraction Fast Point Feature Histogram (FPFH) utilized in the research. However for this project we focus on the robust tracking of known cylindrical landmarks to constrain to the core idea of Extended Kalman Filter Localization.

B. Inertial Measurement

Inertial Measurement Units (IMUs) which is the utilized sensor in the project provides complementary modality to the visual model with high rate proprioceptive data. More specifically gyroscope, accelerometer and a magnetometer which measures objects orientation relative to the Earth’s magnetic North. Due to double integration of sensor data necessity to find position and velocity from the sensor $v_{error} = \int \epsilon dt = \epsilon t$ and $p_{error} = \int \int \epsilon dt dt = \frac{1}{2} \epsilon t^2$ the error increases quadratically resulting in rapid drift even with the slightest noise. The Visual-Inertial Odometry(VIO), a fusion of sensors, seeks to combine strengths of these modalities. A theoretical challenge for VIO mentioned in [4] Zhou et. al 2025 and Yang et. al. 2022, is estimator consistency. Essentially an overconfidence of filter issue due to noisy data resulting in calculation about the states that are theoretically unobservable. The filter calculates a covariance matrix that shrinks rapidly and starts to ignore new sensor measurement that contradicts the current belief. For VIO system the problem stems from ignorance of absolute position and yaw and belief becoming the absolute information. An example for a ground robot is belief of its current height is being zero until robot enters an area of different altitude. When applied “linearization” from EKF, this leads to over confidence and filter divergence. Right invariant method is utilized in both papers. Method treats state as a matrix within Lie Group such as $SE_3(e)$. For estimated state $\hat{x}$ and true state x error is defined as $\delta x^r = \hat{x}x^{-1}$. For full state vector cost of propagation is increased from $O(N)$ to $O(N^2)$. Yang 2022 proposed Decoupled Right Invariant (DRI) method that decouples states from full states and applies certain fixes only in certain feature parts.

C. Error State Extended Kalman Filter

Considering characteristics of both vision model and motion model [2], we adopt the Error State Extended Kalman filter (ES-EKF) framework. The choice of Error State formulation over Full-State EKF is investigated by comparative analysis from Liu et. al. 2019 [10] shows that instead of using full state, computing only error state results 16% increase in processing speed along with increased convergence speed of state and decreased oscillations due to IMU sensor. It should be critical to note that ES-EKF is more sensitive to the faults than the full-state EKF contrasting with the practicality of latent error state space.

In the project we utilize the formulation described by Sola et. al. 2017 [1] which separates the state into "nominal" state and "error" state. Nominal state is integrated from IMU sensor and Error State is estimated from the filter itself. Fundamental principle of Sola's formulation is that the true state x_t is a composition of nominal state x and error state δx resulting in:

$$x_t = x \oplus \delta x$$

Nominal state does not consider noise, it accumulates errors over time. Error state represents the small signal deviation which captures the accumulation of noise and sensor biases. Error state is linearizable therefore it is suitable for the Kalman Filter. Sola utilized unit quaternions for handling 3D rotations which respects the Hamilton convention. While the nominal state uses 4 element (x, y, z, w) to represent full orientation q in S^3 manifold. The error state $\delta\theta$ is a 3 element vector in $\mathbb{R}^3$ tangent space. Mapping between the manifold and the tangent space is handled using exponential and logarithmic maps:

$$\delta q = \exp(\delta\theta) \approx \begin{bmatrix} 1 \\ \frac{1}{2}\delta\theta \end{bmatrix}$$

and true orientation for locally defined errors is derived with quaternion multiplication $q_t = q \otimes \delta q$. The error dynamics are slow-varying compared to non-linear dynamics, as a result it is reported that linearization assumption holds longer durations compared to the standard EKF. While our approach relies on a known map considering the landmark positions. The tight coupling of inertial and visual models principle was proposed in Mourikis et. al. 2007 [3]. Their Multi State Constraint Kalman Filter MSCKF algorithm shows how to fuse high frequency IMU signal with low frequency visual signal from a camera using Error State EKF formulation. MSCKF proposes marginalized features for odometry such as First Estimate Jacobian, Time Delay Compensation and State buffer which will be discussed in later chapters.

D. Zero Velocity Update

In order to prevent error drifts in Inertial Navigation Systems (INS) Zero Velocity Update (ZUPT) is widely implemented method. Kilic et. al. 2023 [11] explains ZUPT as a critical error suppression technique connecting to three benefits of Bounding Velocity Error, Bias Calibration, Limiting Drift Rate. Bounding Velocity Error prevents the velocity estimate from drifting when the robot is stationary. Bias Calibration allows filter to calibrate IMU sensor biases. Kilic reports that ZUPT allows accelerometer and gyroscope bias estimates converge toward the true values and captures true sensor behavior. Limiting Drift Rate effectively limits the rate of position drift that escaped from ES EKF. The paper formulates ZUPT slightly different than pure velocity clamp and includes gyroscope rate:

$$z_{zu} = [-\omega_{IMU}, -v_{neb}]$$

Thus the formulation treats the non-zero gyroscope and accelerometer estimates during stationary robot as the error residual to be minimized.

E. Metrics

Also in this project Absolute Trajectory Error (ATE) also mentioned as Root Mean Square Error (RMSE) along with Normalized Estimation Error Squared (NEES) is utilized. Being widely used metrics in navigation Ning 2022, Zhou 2025, Yang 2022 [5] [4] [9], ATE and NEES serves evaluation for accuracy and consistency of the algorithm.

1) *Absolute Trajectory Error*: Purpose of ATE is accuracy assessment, quantifying how physically close the estimated trajectory is to the ground truth. It evaluates geometric error over the whole trajectory between estimated position (p_{est}) and the ground truth (p_{gt}) at each time step:

$$ATE_{RMSE} = \frac{1}{N} \sum_{k=1}^N ||p_{est,k} - p_{gt,k}||^2$$

2) *Normalized Estimation Error Squared*: Purpose of NEES is consistency assesment which is primary metric for verifying that the filter's uncertainty estimate from matrix P is realistic. Thus comparing if the filter is overconfident or underconfident. Normalizes estimation error by filter's estimated covariance:

$$\epsilon_k = (x_k - \hat{x}_k)^T P_k^{-1} (x_k - \hat{x}_k)$$

Technically ATE should be closer to zero while NEES should be closer to the dimension of the state

II. METHODOLOGY

This study presents a ES-EKF implementation with a visual perception module, responsible from the extraction of the information from camera, and a kinematic controller module, responsible from feasible commands within the kinematic limits.

A. Kinematic Control and Smoothing

The raw velocity commands in a simulation may results with unfeasible physical limits, which may result with unrealistic motions and degrade the IMU performance. We implemented a *Velocity Smoother* module, which acts as a kinematic enforcer, creates continuous smooth velocity profiles from discrete velocity inputs with respect to defined kinematic limits.

We use asymmetric acceleration and deceleration limits, following one of the most common practices of the industry; allowing high deceleration for safety. Moreover, to improve the IMU performance we employ jerk limits, which prevents wheel slip and odometry errors.

B. Visual Perception System

The robot has a monocular camera to detect known cylindrical landmarks with height, $H_{ref} = 1.0m$, and diameter $D_{ref} = 0.4m$. Detection is made based on color-based segmentation and it computes the range and bearing measurements from the detected landmarks.

1) *Image Processing*: The image processing is based on the classical methods, with *OpenCV*, and we are using the HSV color space for robustness against illumination changes of the simulation world. The *visual detector* node creates masks to detect each landmark explicitly. It also applies classical image processing methods, (*dilation* and *erosion*), to eliminate sensor noise. It extracts the contours of the landmarks and checks the sanity of the detection based on the minimum area, $A > 100px$, aspect ratio, < 2.0 , and proximity to image borders to provide stable range and bearing measurements.

2) *Measurement Model*: The *Visual Detector* node uses *Pinhole Camera Model*. The camera intrinsic parameters are derived from the horizontal field of view of the camera,

$$f_x = f_y = \frac{W_{img}}{2 \tan(FOV/2)} \quad (1)$$

$$c_x = \frac{W_{img}}{2} \quad (2)$$

$$c_y = \frac{H_{img}}{2} \quad (3)$$

The *bearing* (ψ) is computed from the centroid u ,

$$\psi = \arctan\left(\frac{c_x - u}{f_x}\right) \quad (4)$$

The *range* (r) is estimated using the observed height of the cylinder h_{pix} , determined from the rotated bounding box to handle tilted projections. The landmarks are projected onto a planar image sensors, thus, objects at the periphery appear stretched due to distortion. The height is corrected using the viewing angle,

$$h_{corr} = h_{pix} \cdot \cos(\psi) \quad (5)$$

Using similar triangles, the range is,

$$r = \frac{H_{ref} \cdot f_y}{h_{corr}} \quad (6)$$

Then the measurement vector $\mathbf{z}_{vis} = [r, \psi]^T$ is given to the filter.

C. Error-State Extended Kalman Filter (ES-EKF)

The ES-EKF is the main estimator of this project. It separates the *large* nominal state from the *low* error state. This approach avoids singularities in the orientation representation.

1) *State Definition*: The **Nominal State** $\mathbf{x} \in \mathbb{R}^{16}$ accumulates the integrated IMU measurements,

$$\mathbf{x} = [\mathbf{p}_W \quad \mathbf{v}_W \quad \mathbf{q}_{WB} \quad \mathbf{b}_a \quad \mathbf{b}_\omega]^T \quad (7)$$

where $\mathbf{p}_W \in \mathbb{R}^3$ is position, $\mathbf{v}_W \in \mathbb{R}^3$ is velocity, $\mathbf{q}_{WB} \in \mathbb{H}$ is the unit quaternion representing *body-to-world* rotation, and $\mathbf{b}_a, \mathbf{b}_\omega \in \mathbb{R}^3$ are accelerometer and gyroscope biases respectively.

The **Error State** $\delta\mathbf{x} \in \mathbb{R}^{15}$ represents the deviation from the true state,

$$\delta\mathbf{x} = [\delta\mathbf{p} \quad \delta\mathbf{v} \quad \delta\boldsymbol{\theta} \quad \delta\mathbf{b}_a \quad \delta\mathbf{b}_\omega]^T \quad (8)$$

where $\delta\boldsymbol{\theta} \in \mathbb{R}^3$ is the rotation vector parameterization of orientation error, avoiding the over-parameterization of quaternion covariance.

2) *IMU Pre-Processing*: The raw IMU measurements are given to a first-order low-pass filter to eliminate noise and provide numerical stability. However, the raw IMU measurements are directly used for stationary detection to accurate motion detection.

3) *Prediction Step*: The nominal state is propagated using bias-corrected IMU measurements,

$$\mathbf{a}_{body} = \mathbf{a}_{filt} - \mathbf{b}_a \quad (9)$$

$$\boldsymbol{\omega}_{body} = \boldsymbol{\omega}_{filt} - \mathbf{b}_\omega \quad (10)$$

$$\mathbf{a}_W = R(\mathbf{q}_{WB}) \cdot \mathbf{a}_{body} + \mathbf{g}_W \quad (11)$$

The state update follows standard kinematic integration,

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \mathbf{v}_k \Delta t + \frac{1}{2} \mathbf{a}_W \Delta t^2 \quad (12)$$

$$\mathbf{v}_{k+1} = \mathbf{v}_k + \mathbf{a}_W \Delta t \quad (13)$$

$$\mathbf{q}_{k+1} = \mathbf{q}_k \otimes \text{Exp}(\boldsymbol{\omega}_{body} \Delta t) \quad (14)$$

4) *Adaptive Process Noise*: The aggressive maneuvers should result with higher uncertainty, so a velocity scaled process noise is employed to provide stability during aggressive maneuvers. Let γ denote the noise scaling factor,

$$\gamma = \begin{cases} 1.0 & \text{if } \|\mathbf{v}_{xy}\| < 0.1 \text{ and } |\omega_z| < 0.05 \\ \gamma_{motion} & \text{if } \|\mathbf{v}_{xy}\| \geq 0.1 \\ \gamma_{motion} \cdot \gamma_{turn} & \text{if additionally } |\omega_z| \geq 0.05 \end{cases} \quad (15)$$

where $\gamma_{motion} = 2.0$ and $\gamma_{turn} = 3.0$. The scaled noise parameters are $\sigma'_a = \gamma\sigma_a$ and $\sigma'_\omega = \gamma\sigma_\omega$.

5) *Error Covariance Propagation*: The error covariance P is propagated using the linearized state transition equation,

$$P_{k+1} = F_x P_k F_x^T + F_i Q_c F_i^T \quad (16)$$

where the Jacobian $F_x \in \mathbb{R}^{15 \times 15}$ captures the error dynamics. The key coupling terms are,

$$F_x[\delta\mathbf{v}, \delta\boldsymbol{\theta}] = -R(\mathbf{q}_{WB})[\mathbf{a}_{body}]_\times \Delta t \quad (17)$$

$$F_x[\delta\mathbf{v}, \delta\mathbf{b}_a] = -R(\mathbf{q}_{WB})\Delta t \quad (18)$$

$$F_x[\delta\boldsymbol{\theta}, \delta\boldsymbol{\theta}] = I - [\boldsymbol{\omega}_{body}]_\times \Delta t \quad (19)$$

$$F_x[\delta\boldsymbol{\theta}, \delta\mathbf{b}_\omega] = -I\Delta t \quad (20)$$

where $[\cdot]_\times$ is the skew-symmetric matrix operator.

6) *Ground Plane Constraints*: This application is for a planar ground vehicle, and it is advantageous to incorporate the planner motion assumption ($z = 0$, $\text{roll} = 0$, $\text{pitch} = 0$). This assumption can be incorporated without introducing pseudo-measurement by applying it as a hard constraint after each prediction step. In the end the nominal state components are directly resets,

$$p_z \leftarrow 0, \quad v_z \leftarrow 0 \quad (21)$$

$$\mathbf{q} \leftarrow \text{Euler}(0, 0, \text{yaw}(\mathbf{q})) \quad (22)$$

The corresponding covariance elements are set to almost zero, not zero for avoiding singularity, and, all cross-correlations with constrained states are zeroed to maintain filter consistency,

$$P_{ij} = P_{ji} = 0, \quad P_{ii} = \epsilon \quad \forall i \in \{z, v_z, \phi, \theta\} \quad (23)$$

7) *Stationarity Detection and ZUPT*: Velocity drift is almost certain with IMU measurement integration. Thankfully, it can be corrected with Zero velocity Updates (ZUPT). First the vehicle acceleration is determined,

$$\mathbf{a}_{kin} = (\mathbf{a}_{raw} - \mathbf{b}_a) + R(\mathbf{q}_{WB})^T \mathbf{g}_W \quad (24)$$

The vehicle is stated as stationary if the following condition holds,

$$\begin{aligned} \|\mathbf{a}_{kin,xy}\| < \tau_{acc} \quad \wedge \quad |\omega_{z,raw} - b_{\omega,z}| < \tau_{gyro} \quad (25) \\ \wedge \quad \|\mathbf{v}_{xy}\| < \tau_{vel} \quad (26) \end{aligned}$$

for N consecutive samples. A safety guard is implemented to detect stationary, where repeated checks are required to be accepted as stationary.

If vehicle is stationary, the filter fuses a pseudo-measurement $\mathbf{z}_{zupt} = \mathbf{0}$ for the velocity,

$$H_{zupt} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & 1 & 0 & \dots \end{bmatrix} \quad (27)$$

Post-update, a covariance floor is applied to prevent over-confidence, $P_{v_x v_x}, P_{v_y v_y} \leq 4R_{zupt}$.

8) *Visual Measurement Update*: The visual measurements are incorporated as in the standard EKF update. The visual measurements are taken from the known landmarks, and given a known landmark position $\mathbf{l}_W$ in the world frame, the measurement at the camera frame,

$$\mathbf{l}_{body} = R(\mathbf{q}_{WB})^T (\mathbf{l}_W - \mathbf{p}) \quad (28)$$

$$\mathbf{l}_{cam} = \mathbf{l}_{body} - \mathbf{t}_{cam} \quad (29)$$

where $\mathbf{t}_{cam}$ is the camera offset in the body frame. The predicted measurements are,

$$\hat{r} = \|\mathbf{l}_{cam,xy}\| \quad (30)$$

$$\hat{\psi} = \arctan 2(l_{cam,y}, l_{cam,x}) \quad (31)$$

The innovation is $\mathbf{y} = [r - \hat{r}, \text{wrap}(\psi - \hat{\psi})]^T$.

The measurement Jacobian $H \in \mathbb{R}^{2 \times 15}$ is computed analytically. Let x_c, y_c be the landmark coordinates in the camera frame,

$$\frac{\partial r}{\partial \mathbf{l}_{cam}} = \frac{1}{r} [x_c, y_c, 0]^T \quad (32)$$

$$\frac{\partial \psi}{\partial \mathbf{l}_{cam}} = \frac{1}{r^2} [-y_c, x_c, 0]^T \quad (33)$$

These are chained through,

$$H[\cdot, \delta \mathbf{p}] = \frac{\partial h}{\partial \mathbf{l}_{cam}} \cdot (-R^T) \quad (34)$$

$$H[\cdot, \delta \boldsymbol{\theta}] = \frac{\partial h}{\partial \mathbf{l}_{cam}} \cdot [\mathbf{l}_{body}]_{\times} \quad (35)$$

Then the Kalman update,

$$S = HPH^T + R_{vis} \quad (36)$$

$$K = PH^T S^{-1} \quad (37)$$

$$\delta \mathbf{x} = K\mathbf{y} \quad (38)$$

$$P \leftarrow (I - KH)P(I - KH)^T + KR_{vis}K^T \quad (39)$$

9) *Error Injection and Reset*: The error state correction is injected into the nominal state,

$$\mathbf{p} \leftarrow \mathbf{p} + \delta \mathbf{p} \quad (40)$$

$$\mathbf{v} \leftarrow \mathbf{v} + \delta \mathbf{v} \quad (41)$$

$$\mathbf{q} \leftarrow \mathbf{q} \otimes \text{Exp}(\delta \boldsymbol{\theta}) \quad (42)$$

$$\mathbf{b}_a \leftarrow \mathbf{b}_a + \delta \mathbf{b}_a \quad (43)$$

$$\mathbf{b}_{\omega} \leftarrow \mathbf{b}_{\omega} + \delta \mathbf{b}_{\omega} \quad (44)$$

Following orientation injection, the covariance is rotated to align with the new nominal,

$$P \leftarrow GPG^T, \quad G_{[\delta \boldsymbol{\theta}, \delta \boldsymbol{\theta}]} = I - \frac{1}{2}[\delta \boldsymbol{\theta}]_{\times} \quad (45)$$

with $G = I_{15}$ elsewhere. Ground constraints are re-applied after each injection.

III. SIMULATION SETUP

A. Software Environment

The simulation environment was developed using ROS2 Humble and Gazebo Fortress (Ignition Gazebo). The system architecture consists of five main ROS2 nodes,

- **Visual Detector**: Detects landmarks and extracts information with respect to landmarks.
- **ES-EKF**: Estimates state of the vehicle.
- **Trajectory Publisher**: Generates predefined motion patterns.
- **Velocity Smoother**: Ensures the passed commands to the vehicle respects to the kinematic limits.
- **Trajectory Monitor**: Gathers the data from system and publishes the performance metrics.
- **Data Logger**: Saves the system metrics and measurements for data visualization.

Communication between Gazebo and ROS2 was handled through the `ros_gz_bridge` package, which translates Ignition messages to ROS2 message types for IMU data, camera images, velocity commands, and simulation clock.

B. Robot Platform

The robot platform is a well known type of *turtlebot* which is an differential drive ground vehicle equipped with an IMU sensor and a forward-facing monocular camera. The IMU operates at approximately 50 Hz with configurable Gaussian noise parameters for both accelerometer and gyroscope measurements. The camera has a resolution of 640×480 pixels with a horizontal field of view of 60 and operates at 30 Hz. The camera is mounted at a height of 0.4 m from the ground, facing forward along the robot's heading direction.

C. Landmark Environment

The simulation environment contains 10 cylindrical landmarks with distinct colors distributed throughout the workspace. Each landmark consists of a static cylinder with radius 0.2 m and height 1.0 m. Landmark colors were generated using hue distribution to ensure maximum visual separation between landmarks. A Python script generates both the Gazebo world file (SDF format) and a corresponding

JSON file containing landmark positions and HSV color ranges for detection.

The visual detector node performs color-based landmark detection using HSV color space segmentation. For each landmark, a ± 5 hue range around the nominal hue value is used, with saturation and value thresholds set to reject background pixels. Range estimation is performed using the pinhole camera model based on the apparent pixel height of detected landmarks, while bearing is computed from the horizontal pixel offset relative to the image center.

D. State Estimation

The ES-EKF maintains a 15-dimensional error state vector comprising position error (3), velocity error (3), orientation error (3), accelerometer bias (3), and gyroscope bias (3). The nominal state is propagated using IMU measurements while the error state covariance is updated through the standard EKF prediction equations. Visual landmark measurements provide range and bearing observations that are fused through the EKF measurement update step.

Ground truth poses are obtained from the Gazebo simulation through the pose information topic, which provides the true robot position and orientation at each timestep. These ground truth values are used exclusively for performance evaluation and are not available to the filter during operation.

E. Experimental Trajectories

Two trajectory patterns were used to evaluate the system. The figure-8 trajectory consists of two circular loops of radius 3 m traversed at a linear velocity of 0.5 m/s, resulting in a total path length of approximately 37.7 m over 90 seconds. The square trajectory follows a 6×6 m square path at the same velocity, with 90° turns at each corner, covering approximately 24 m over 150 seconds including turn durations. Both trajectories were executed using open-loop velocity commands

IV. EXPERIMENTAL RESULTS

The proposed visual-inertial navigation system was evaluated through simulation experiments using two distinct trajectory patterns: a figure-8 shape and a square path. Each experiment was conducted with 10 color-coded landmarks distributed throughout the environment. Performance was assessed using Root Mean Square Error (RMSE), Normalized Estimation Error Squared (NEES), and instantaneous position error metrics. Two sensor configurations were compared: IMU-only dead reckoning and fused IMU-vision estimation.

A. IMU-Only Navigation Performance

The IMU-only configuration gives a drift which was already expected for a dead reckoning system. On the figure-8 trajectory, the filter accumulated a final RMSE of 9.81 m over approximately 100 seconds of operation, with position error growing monotonically to approximately 17.5 m, as shown in Fig. 1. The square trajectory showed similar degradation, reaching a final RMSE of 13.04 m with maximum position errors exceeding 20 m (Fig. 3).

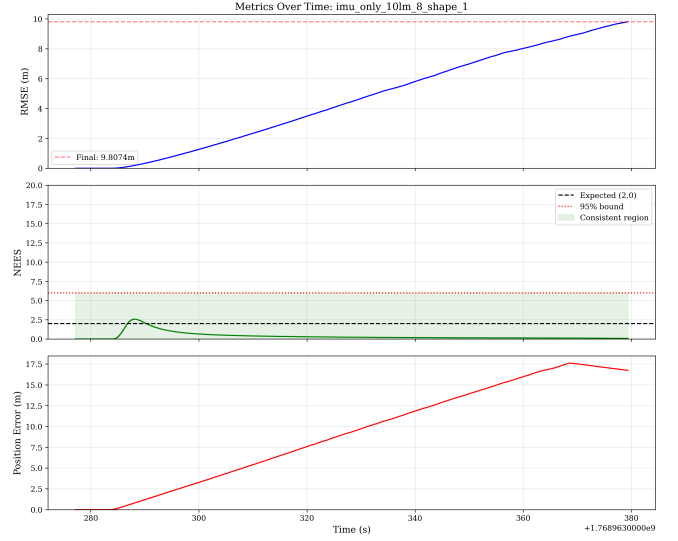


Fig. 1: Performance metrics for IMU-only navigation on the figure-8 trajectory showing RMSE, NEES, and position error over time.

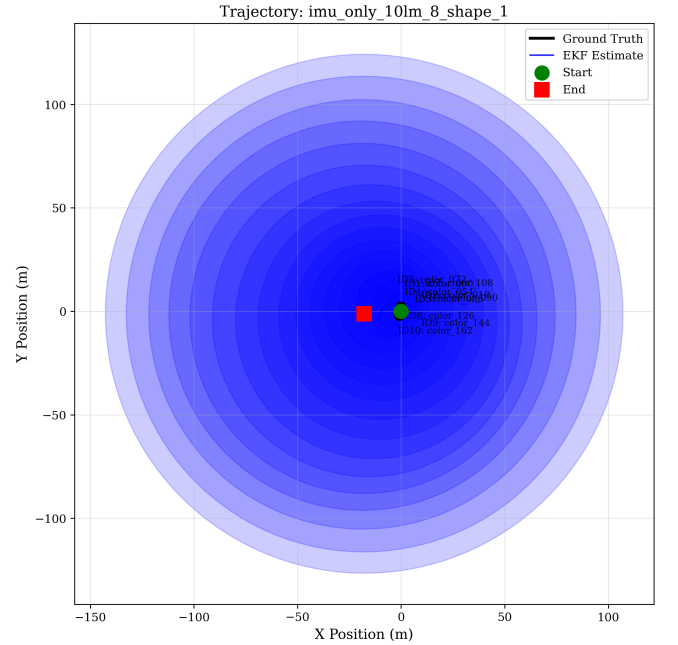


Fig. 2: Trajectory comparison for IMU-only navigation on the figure-8 path. The rapidly expanding uncertainty ellipses illustrate unbounded covariance growth characteristic of dead reckoning.

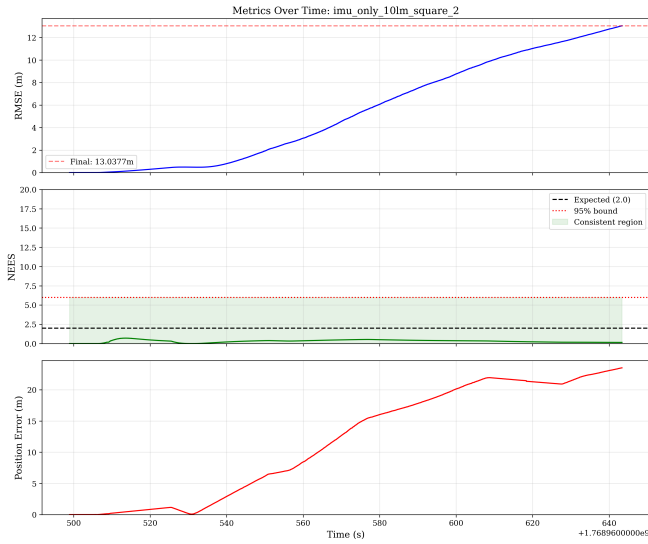


Fig. 3: Performance metrics for IMU-only navigation on the square trajectory.

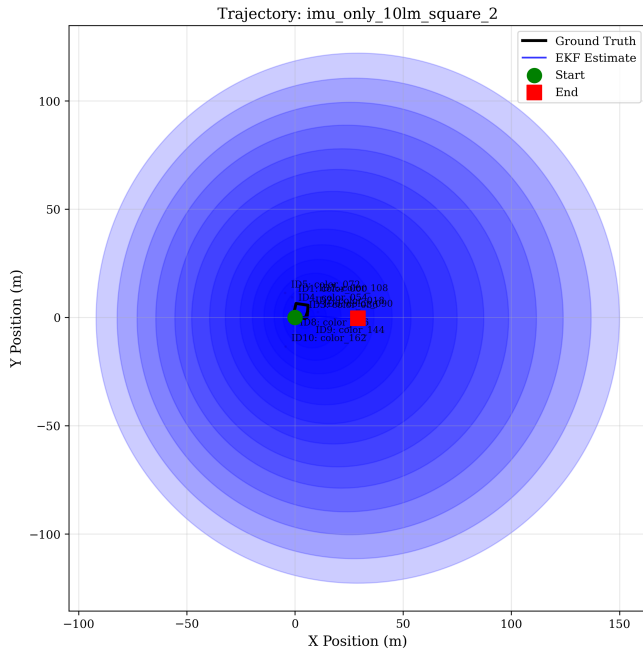


Fig. 4: Trajectory comparison for IMU-only navigation on the square path showing severe drift accumulation.

The NEES values for IMU-only estimation stayed below the expected 2.0, which indicates an overconfident filter. This shows up in the trajectory plots (Figs. 2 and 4), where the uncertainty ellipses expand rapidly, covering hundreds of meters, while the actual position error stays relatively small. The covariance estimates do capture the growing uncertainty when there are no external corrections, but the NEES suggests that the process noise parameters might need to be adjusted.

B. Visual-Inertial Fusion Performance

Integration of visual landmark measurements significantly improved localization accuracy. Figure-8 trajectory shows that our method has achieved a final RMSE of 0.70 m (Fig. 5), proves a 93% reduction compared to IMU-only navigation. The square trajectory showed a final RMSE of 2.49 m (Fig. 7), corresponding to an 81% improvement over dead reckoning.

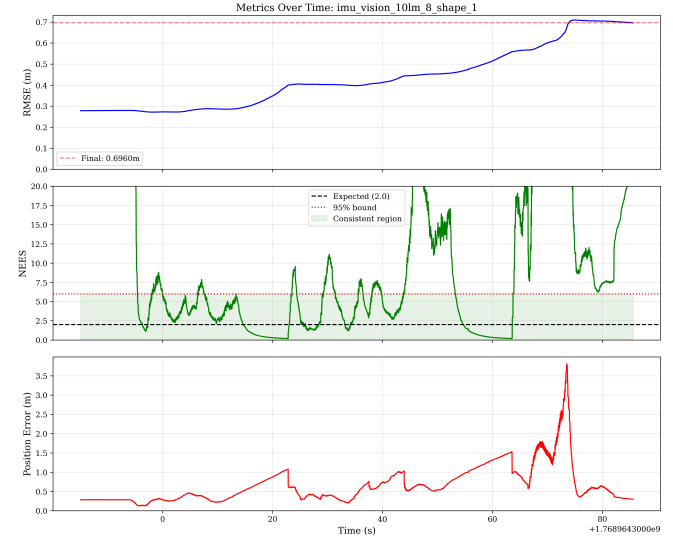


Fig. 5: Performance metrics for visual-inertial fusion on the figure-8 trajectory demonstrating bounded error growth.

From the trajectory plots in Figs. 6 and 8, it can be seen that adding visual measurements helps reduce the error in the estimation. In both experiments, the EKF result follows the ground truth more closely than the IMU-only case. The uncertainty ellipses in the fused setup are also smaller, which shows that the landmark observations help correct the drift coming from the IMU.

C. Filter Consistency Analysis

For the figure-8 trajectory (Fig. 5), the NEES value goes above 15 at several points. In the square trajectory case (Fig. 7), NEES stays high for longer periods, mostly between 10 and 20. This suggests that the filter is too confident compared to the actual error. Such behavior is common in EKF-based visual-inertial systems and is usually caused by linearization effects and possible inaccuracies in the measurement model.

At these points, there are no visible landmarks. Since no measurement update is performed and the filter runs only with IMU data, the error increases and the NEES value becomes high.

D. Landmark Detection Statistics

Visual measurement statistics are presented in Figs. 9 and 10. The detection data indicate non-uniform landmark observability across both trajectories. On the figure-8 path, landmark ID3 received the highest detection count (approximately 390 detections), while other landmarks were observed

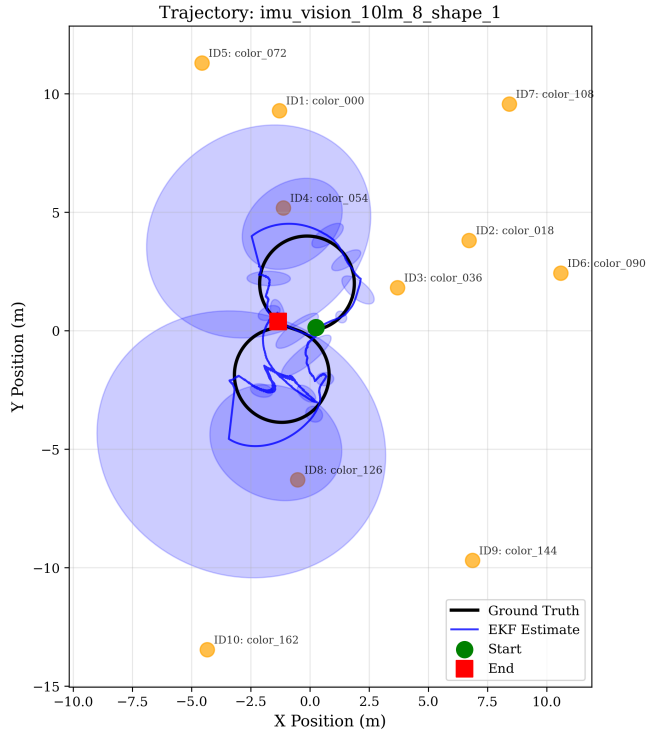


Fig. 6: Trajectory comparison for visual-inertial fusion on the figure-8 path. Orange markers indicate landmark positions with their color IDs. Uncertainty ellipses remain bounded due to visual corrections.

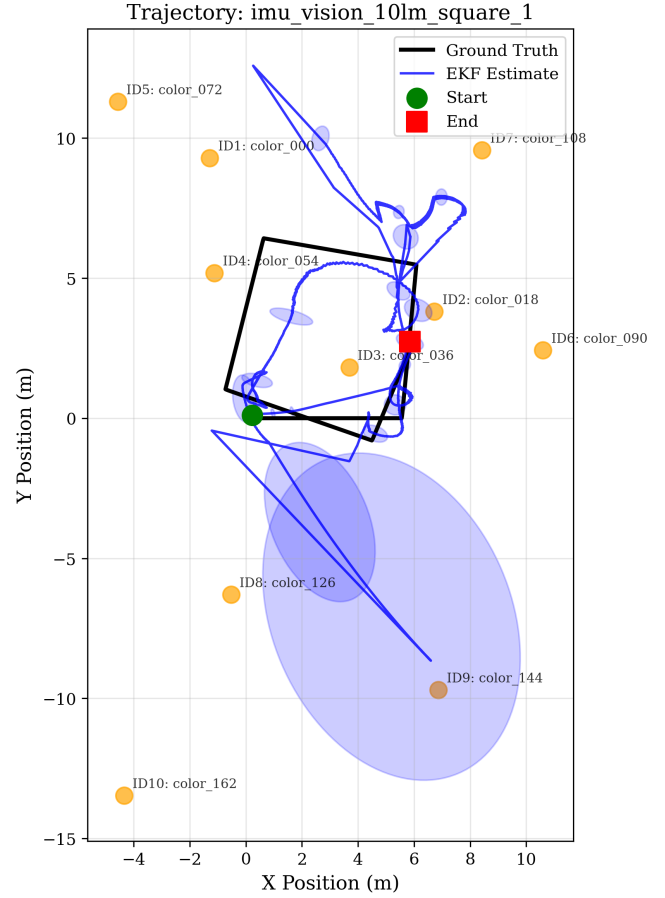


Fig. 8: Trajectory comparison for visual-inertial fusion on the square path showing improved tracking accuracy compared to IMU-only navigation.

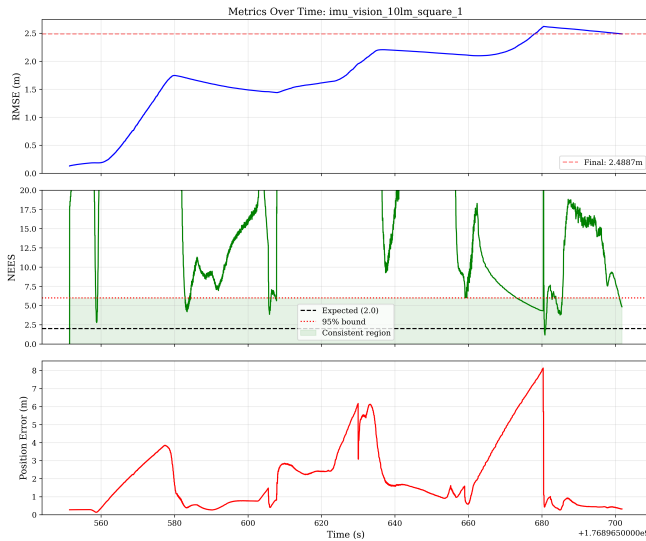


Fig. 7: Performance metrics for visual-inertial fusion on the square trajectory.

between 130-250 times depending on their positions relative to the trajectory. The square trajectory showed landmarks ID6 and ID7 as most frequently observed (approximately 350 detections each), with landmark ID9 receiving notably fewer observations (approximately 40 detections).

Range measurements were distributed primarily between 2-16 m for the figure-8 trajectory and 2-20 m for the square path, with bearing angles spanning the full $\pm 30^\circ$ field of view. The detection distributions reflect the geometric relationship between the robot's path and landmark positions, with centrally-located landmarks (ID3, ID4, ID8) receiving more consistent observations across both trajectory types.

E. Summary

Table I summarizes the quantitative performance comparison between the two sensor configurations.

TABLE I: Performance Comparison: IMU-Only vs. Visual-Inertial Fusion

Traj.	Config.	RMSE (m)	Max (m)	Impr.
Fig-8	IMU	9.81	17.5	–
	IMU+Vis	0.70	3.7	93%
Square	IMU	13.04	23.0	–
	IMU+Vis	2.49	8.1	81%

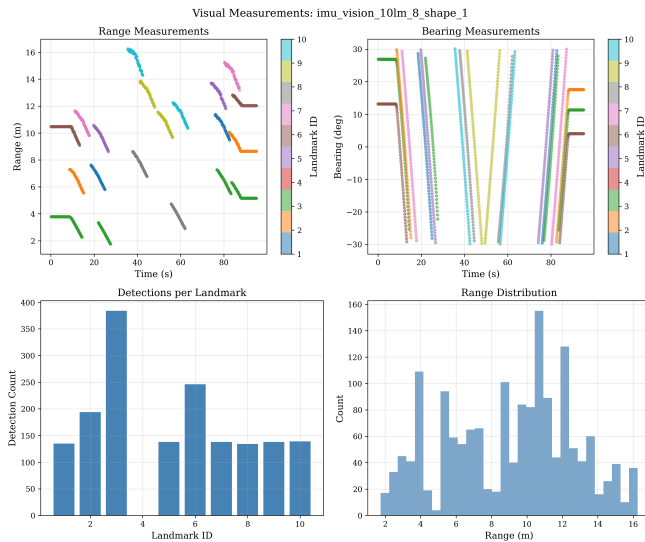


Fig. 9: Visual measurement statistics for the figure-8 trajectory: range and bearing measurements over time (top), detection counts per landmark, and range distribution (bottom).

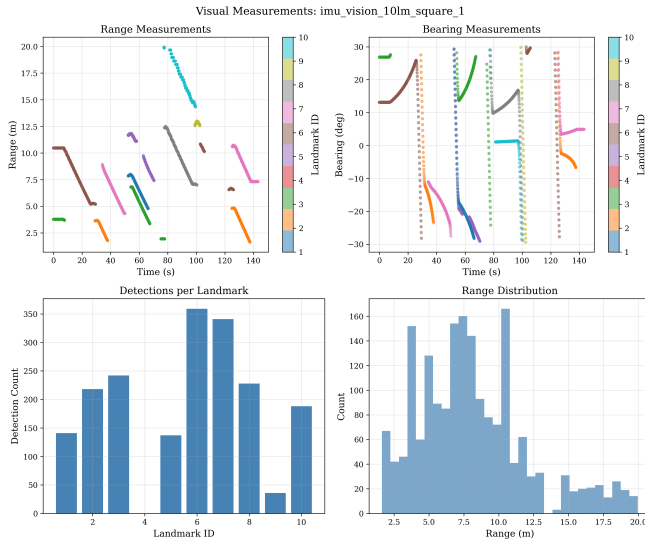


Fig. 10: Visual measurement statistics for the square trajectory showing landmark detection patterns and measurement distributions.

V. DISCUSSION

A. Performance Analysis

The experimental results clearly shows the expected results on using the inertial and visual sensing for robot localization. The IMU provides high-frequency state propagation but suffers from unbounded drift due to the double integration of noisy acceleration measurements. The visual system offers absolute position corrections through landmark observations but is limited by field of view constraints and detection range. The fusion of both modalities through the ES-EKF achieves better performance.

The difference in accuracy between the two trajectory types needs another examination. The figure-8 trajectory

achieved a final RMSE of 0.70 m compared to 2.49 m for the square trajectory. Reason for this change can be considered to several factors. The figure-8 path maintains continuous curved motion, providing consistent observability of the orientation state through gyroscope measurements. In contrast, the square trajectory involves extended straight segments followed by stationary 90° turns, during which the robot briefly stops and rotates in place. These discrete motion phases create periods where certain state components become less observable, allowing error to accumulate before the next visual correction. Also due to the error on robot command, linear motion drifts more than circular one.

Additionally, the landmark distribution relative to each trajectory affects correction frequency. The figure-8 trajectory passes through the central region of the workspace where landmarks are more densely distributed, resulting in more frequent visual updates. The square trajectory extends to the workspace periphery where fewer landmarks are visible, leading to longer intervals between corrections and thus larger drift in position.

B. Filter Consistency

The NEES analysis shows the difference between filter's estimated uncertainty and actual estimation error. A consistent filter should produce NEES values distributed around the state dimension (2 for the 2D position subspace being evaluated) with 95% of values falling below approximately 6.0. The observed NEES values frequently exceeded 15, indicating that the filter is overconfident in its estimates.

Several factors contribute to this inconsistency. First, the EKF linearization introduces errors when the system operates far from the linearization point. During rapid maneuvers or when large corrections are applied, this approximation degrades. Second, the process noise covariance matrix Q may not accurately reflect the true uncertainty in the IMU measurements and motion model. Tuning Q involves a trade-off: smaller values produce tighter uncertainty bounds but risk inconsistency, while larger values maintain consistency at the cost of increased estimation variance.

The measurement noise covariance R also influences consistency. The range estimation method, which detects distance from the pixel height of landmarks, introduces errors that depend on detection quality and partial occlusions. A fixed R value cannot capture this state-dependent uncertainty, leading to periods where the filter trusts more than usual to visual measurements. Apart from these we realized when there is no measurement, this values peak.

Interestingly, the IMU-only configuration exhibited the opposite behavior, with NEES values consistently below the expected value. This indicates an overly conservative filter that correctly recognizes growing uncertainty but propagates covariance faster than errors actually accumulate. While this is preferable to overconfidence from a safety perspective, it suggests the process noise parameters could be reduced for tighter uncertainty bounds when visual corrections are unavailable.

C. Landmark Detection Performance

The HSV-based color detection approach performed reliably within the synthetic Gazebo environment. The hue distribution ensured sufficient color separation between landmarks, eliminating false associations that could corrupt the filter state. That is intentionally applied by our scene generation script. Detection rates varied across landmarks depending on their positions relative to the trajectory, with centrally located landmarks receiving up to three times more observations than peripheral ones.

The range estimation accuracy depends critically on accurate detection of landmark boundaries. Partial occlusions, image noise, or lighting variations can cause the detected bounding box to differ from the true landmark extent, introducing range measurement errors. The bearing measurement, derived from horizontal pixel position, proved more robust as it depends only on the centroid location rather than boundary detection.

The limited camera field of view ($\pm 30^\circ$) constrains the number of simultaneously visible landmarks. During straight-line segments where the robot heading remains constant, only landmarks directly ahead contribute corrections. Turns temporarily bring additional landmarks into view which lowered the drift.

D. Limitations

There are several limitations in the setup. The requirement for known landmark positions restricts deployment to pre-mapped environments. The color-based detection assumes controlled lighting conditions that may not hold in outdoor or variable illumination scenarios. The 2D ground plane assumption, while appropriate for the differential drive platform used here, would require modification for aerial vehicles or uneven terrain.

The filter tuning was performed manually through iterative adjustment of Q and R matrices. While functional, this approach does not guarantee optimal performance and may not generalize across different trajectories or noise conditions. Adaptive methods that estimate noise parameters online could improve robustness to varying operating conditions.

Finally, the evaluation was conducted entirely in simulation where ground truth is perfectly known and sensor models are idealized. Real-world deployment would introduce additional challenges including sensor calibration errors, time synchronization between IMU and camera, and environmental factors such as motion blur and dynamic lighting.

VI. CONCLUSION

This work presented a visual-inertial navigation system based on the Error State Extended Kalman Filter for ground robot localization. The system fuses IMU measurements with monocular camera observations of color-coded landmarks to achieve bounded position estimation error.

Experimental evaluation demonstrated the critical importance of visual corrections for maintaining localization accuracy. IMU-only dead reckoning resulted in unbounded drift, with final RMSE values of 9.81 m and 13.04 m for the

figure-8 and square trajectories respectively. The integration of visual landmark measurements reduced these errors to 0.70 m and 2.49 m, representing improvements of 93% and 81% over pure inertial navigation.

The NEES analysis revealed that the filter exhibits periods of inconsistency, with values frequently exceeding the 95% confidence bound. This suggests that the linearization errors inherent to the EKF formulation, combined with potential inaccuracies in the measurement noise model, result in optimistic covariance estimates. Future work should address this through improved noise characterization or alternative filtering approaches such as the Unscented Kalman Filter.

The color-based landmark detection approach proved effective in the synthetic Gazebo environment, with reliable detection across varying ranges from 2 m to 16 m. However, the non-uniform landmark observability observed in both trajectories highlights the importance of landmark placement for consistent filter performance. Landmarks positioned closer to the trajectory center received significantly more observations.

Several directions remain for future development. First, extending the system to handle unknown landmark positions through simultaneous localization and mapping would eliminate the requirement for prior environment knowledge. Second, adding additional sensor modalities such as wheel odometry could provide velocity constraints during periods of limited visual observations. Finally, real-world deployment would require more robust landmark detection methods capable of handling lighting variations and partial occlusions not present in simulation.

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